KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE HEALTH SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRICAL ENGINEERING TECHNOLOGY THEORY

DATE: 30th September, 2022	**TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

SECTION 1 - MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. Which of the following statements is NOT true about atoms?
   a) A negative body will attract electrons
   b) A positive body will attract electrons
   c) Neutrons have no charge
   d) Number of electrons is equal to the number of protons in a balanced atom
2. E.M.F. is induced by which principle of electromagnetism?
   a) By using a small cross-sectional area coil
   b) By increasing the size of the core
   c) By increasing the strength of the magnetic field
   d) By changing the direction of rotating the conductor
3. According to Kirchoff's first law
   a) Currents flowing towards a junction from opposites will cancel one another
   b) Current going to each branch of a parallel circuit will divide equal
   c) Following the second Kirchoff's Law the voltage sum of voltage drops in a series circuit will be same in all the resistors connected
   d) Voltage can be measured using an integrated ohmmeter frequency meter in a circuit
4. Which one among the following is NOT an effect of electricity on the human body?
   a) Burns on the skin
   b) Causes an irregular heart beat
   c) Relaxes the body muscles
   d) Causes an increase in the blood flow
5. Which of the following is NOT a form of an energy in an AC circuit?
   a) Constant power
   b) Reactive power
   c) Real power
   d) Apparent power
6. Which one of the following is NOT a factor determining the magnitude of magnetic field?
   a) Magnetic field density (B)
   b) Orientation with respect to surface area
   c) The area of the surface (A)
   d) The manner in which windings are mounted on the core
7. When current flows through an ideal conductor?
   a) The conductor resistance increases
   b) The conductor resistance decreases
   c) A magnetic field is created around the conductor
   d) The conductor becomes heated up
8. Which one among the following is NOT a force acting on the pointer of a measuring instrument?
   a) Deflecting force
   b) Centrifugal force
   c) Damping force
   d) Controlling force

9. Which of the following is NOT true about effects of armature reaction in DC machines?
   a) Distorts the main flux
   b) Weakens the main flux
   c) Shifts the Magnetic Neutral Axis
   d) Reduces the generated voltage
10. The following equations are true about DC machines. Which one is NOT correct?
   a) $V = E_b + I_a R_a$
   b) $V = E_b - I_a R_a$
   c) $E_g = V + I_a R_a$
   d) $I_a = \frac{V - E_b}{R_a}$
11. Which one of the following is NOT a contributing factor to losses in DC generators?
   a) Frequency
   b) Flux density
   c) Volume and type of core
   d) The resistance of coil
12. Which of the following is NOT an advantage of flux control method used to control speed of a DC motor?
   a) Flux can be increased only up to its rated value
   b) It provides relatively smooth and easy control
   c) Speed control above rated speed is possible
   d) As the field winding resistance is high the field current is small
13. The speed of a 4 pole, 60HZ motor in RPM is
   a) 2400
   b) 1800
   c) 1200
   d) 1500
14. Which of the following is NOT true about shaded pole motors?
   a) Shaded pole motors produces very small starting torque
   b) The efficiency of a shaded pole motor is low
   c) Used for small fans and small appliances
   d) The direction of rotation of s shaded pole motor cannot be changed
15. Which of the following motors can be connected to either DC or AC supply.
   a) Repulsion motor
   b) Reluctance motor
   c) Hysteresis motor
   d) Universal motor
16. Which of the following is not true about a star connected three-phase AC generator?
   a) Line voltage = $\sqrt{3}$ x phase voltage
   b) Line current = phase current
   c) Line voltages are 120° apart
   d) Line voltages are 30° ahead of respective phase voltages
17. The direction of rotation of a DC series motor can be changed by
   a) Interchanging supply terminals
   b) Interchanging field terminals
   c) Either A or B
   d) None of the above

18. Which type of rotor is used for low speed AC generators?
    a) Cylindrical rotor
    b) Salient pole rotor
    c) Squirrel cage
    d) Armature type
19. Which of the following is not true about windings connection in three-phase AC generators
    a) The windings are connected through slip rings in bigger generators
    b) The windings are connected fixed terminals using studs
    c) Large cable sizes are used to generate high voltages
    d) In order to overcome insulation resistance problem, the armature is kept stationary for large generators
20. Rotor slots of a three-phase induction motor are skewed to
    a) Increase efficiency
    b) Reduce humming
    c) Support magnetic locking of stator and rotor
    d) Provide torque to the rotor
21. Wave winding is composed of
    a) An even number of conductors
    b) An odd number of conductors
    c) An even number which is exact multiple of poles + 2
    d) An even number which is exact multiple of poles
22. Copper losses in a transformer are measured using
    a) Closed circuit current
    b) Open circuit current
    c) Both (a) and (b)
    d) None of the above
23. Which of the following is an advantage of a transformer?
    a) It is a stationary device
    b) It provides high efficiency
    c) No starting time is required
    d) Operates both on ac and dc
24. A transformer connected in star fashion works at _____ condition.
    a) No load
    b) Full load
    c) Balanced and unbalanced load
    d) Balanced load
25. The efficiency of an induction motor is about
    a) 100%
    b) 80-90%
    c) 50-60%
    d) Less than 50%

26. It is advisable to avoid DOL line-starting of induction motor and use starters because
    a) Motor takes five to seven times its full load current
    b) It will pick-up very high speed and may go out of step
    c) It will run in reverse direction
    d) Starting is very high
27. In three-phase squirrel cage induction motors
    a) Rotor conductor ends are short-circuited through slip rings
    b) Rotor conductors ae short-circuited through end rings
    c) Rotor conductors are kept open
    d) Rotor conductors are connected to insulation
28. In a three-phase induction motor, the number of poles in the rotor winding is
    a) Zero
    b) More than the number of poles in stator
    c) Less than the number of poles in stator
    d) Equal to number of poles in stator
29. The basic function of a circuit breaker is to
    a) Maintain the arc
    b) Extinguish the arc
    c) Interrupt excess current
    d) Emit the ionization electrons
30. What is the main purpose of submerging contacts of high voltage switches in oil?
    a) Insulate the contacts from switch body
    b) Lubricate the contacts
    c) Conduct current through contacts
    d) Extinguish the arc
31. Which one of the following devices is used for power factor correction in substations?
    a) Induction motors
    b) Slip ring induction motor
    c) DC series generator
    d) Synchronous condenser
32. In a pure capacitor
    a) Voltage leads current by 30^0
    b) Voltage leads current by 90^0
    c) Current and voltage are in phase
    d) Current leads Voltage by 90^0
33. The frequency of an alternating current is
    a) The speed with which the alternator runs
    b) The number of cycles generated in one minute
    c) The number of waves passing through a point in one second
    d) The number of electrons passing through a point in one second
34. Alternating current cannot be measured by DC ammeter, because
    a) AC is virtual
    b) AC changes in direction
    c) AC cannot pass through dc ammeter
    d) Average value of AC for completer cycle is zero

35. In a purely resistive circuit
    a) Current leads the voltage
    b) Current lags behind voltage
    c) Current and voltage are in phase
    d) An of the above may be true depending upon the value of resistance
36. The capacitance of the parallel capacitor increases when
    a) Area of the plate is decreased
    b) Area of the plate is increased
    c) Distance between the plates is increased
    d) Thickness of the plate is increased
37. A surface that has same electrostatic potential at every point on it is known as
    a) Equal-potential surface
    b) Same potential surface
    c) Equal magnitude surface
    d) Equipotential surface
38. In homes, electrical appliances are connected
    a) In parallel with source only if it is a high power appliance
    b) In series with the source
    c) Some in series and some in parallel with source
    d) In parallel with the source
39. The dielectric loss of a capacitor can be measured by which of the following?
    a) Megger
    b) Schering bridge
    c) Potentiometer
    d) Loss of charge method
40. Which of the following measuring instrument is odd?
    a) Recording instrument
    b) Integrating instrument
    c) Display instrument
    d) Indicating instrument

SECTION 2 – SHORT ANSWER QUESTIONS (40 MARKS)

41. (a) State three factors that affect the capacitance of a capacitor. (3 marks)
    (b) State the two factors considered when selecting a capacitor. (2 marks)
42. Determine the capacitance of a capacitor with 5 plates each of area 400cm^2 separated by a dielectric of thickness 0.25mm, whose relative permittivity is 5.
    Take $\epsilon_o = 8.85 \times 10^{-12}$. (5 marks)
43. What is the function of the following devices incorporated in a motor protection switch gear?
    (a) Thermal Trip (2 ½ marks)
    (b) Magnetic Trip (2 ½ marks)
44. (a) State three factors that determine the speed of a DC motor? (3 marks)
    (b) How can armature reaction be minimized in DC motors? (2 marks)
45. List five types of single phase induction motors. (5 marks)
46. (a) State the two types of 3-phase induction motors in common use. (2 marks)

(b) Draw diagrams of star and delta windings as used in 3-phase induction motors. (3 marks)

47. (a) State three main losses that take place in a three-phase transformers. (3 marks)

(b) Which principle does a transformer use to induce voltage? (1 mark)

(c) State the method used to fix the primary and secondary windings on a transformer core (1 mark)

48. (a) The core of DC electromagnets used in generators is laminated to reduce (1 mark)

(b) State four prime movers used to drive the alternator. (4 marks)

SECTION 3 – LONG ANSWER QUESTION (20 MARKS)

49. (a) What is the difference between lap and wave windings? (4 marks)

(b) Why are slip-rings used in AC generators? (1 mark)

(c) A three phase 16 pole alternator has a star connected winding with 144 slots and 10 conductors per slot. The flux per pole is 30mW sinusoidaly distributed and the speed is 375 rpm. Winding factor = 1. Determine:

i. The frequency and phase voltage. (6 marks)

ii. The line voltage. (2 marks)

(d) State three ideal characteristics of a transformer. (3 marks)

(e) State the effect of the following on 3-phase induction motors:

i. Over-voltage (2 marks)

ii. Under voltage (2 marks)